Title :- BST construction from Given element

Suggested modifications:
1) Display the tree after each insertion
2) Calculate the height and leaf count of the BST
3) Allow the user to search for and delete a selected value.

problem Statement:- Write a c++ program to construct a Binary Search tree (BST) using the elements 14, 15, 4, 9, 7, 18, 3 and 5 in the given order display the tree after each insertion and perform all three depth first traversal, inorder, preorder and postorder. calculate the height and total numbers of leaf nodes of the final BST. Also allow the user to search for and delete a selected element from the BST

objectives
1) To construct a Binary Search tree using the given elements.
2) To display the BST after each insertion
3) To perform inorder, preorder, and postorder traversals.
4) To calculate the height and number of leaf nodes in the BST.
5) To search for a specified element in the BST
6) To delete a selected element while maintaining the BST properties.

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Algorithm

Step 1: Start
Step 2: Initialize the root of the Binary Search tree (BST) as NULL.
Step 3: insert the elements 14, 15, 4, 9, 7, 18, 3, into BST in the give order.

Step 4: After each insertion, display the current BST structure.

Step 5: perform the following depth-first traversals.
- inorder: traverse left → Root → right
- preorder: traverse Root → left → right
- postorder: traverse left → right → Root

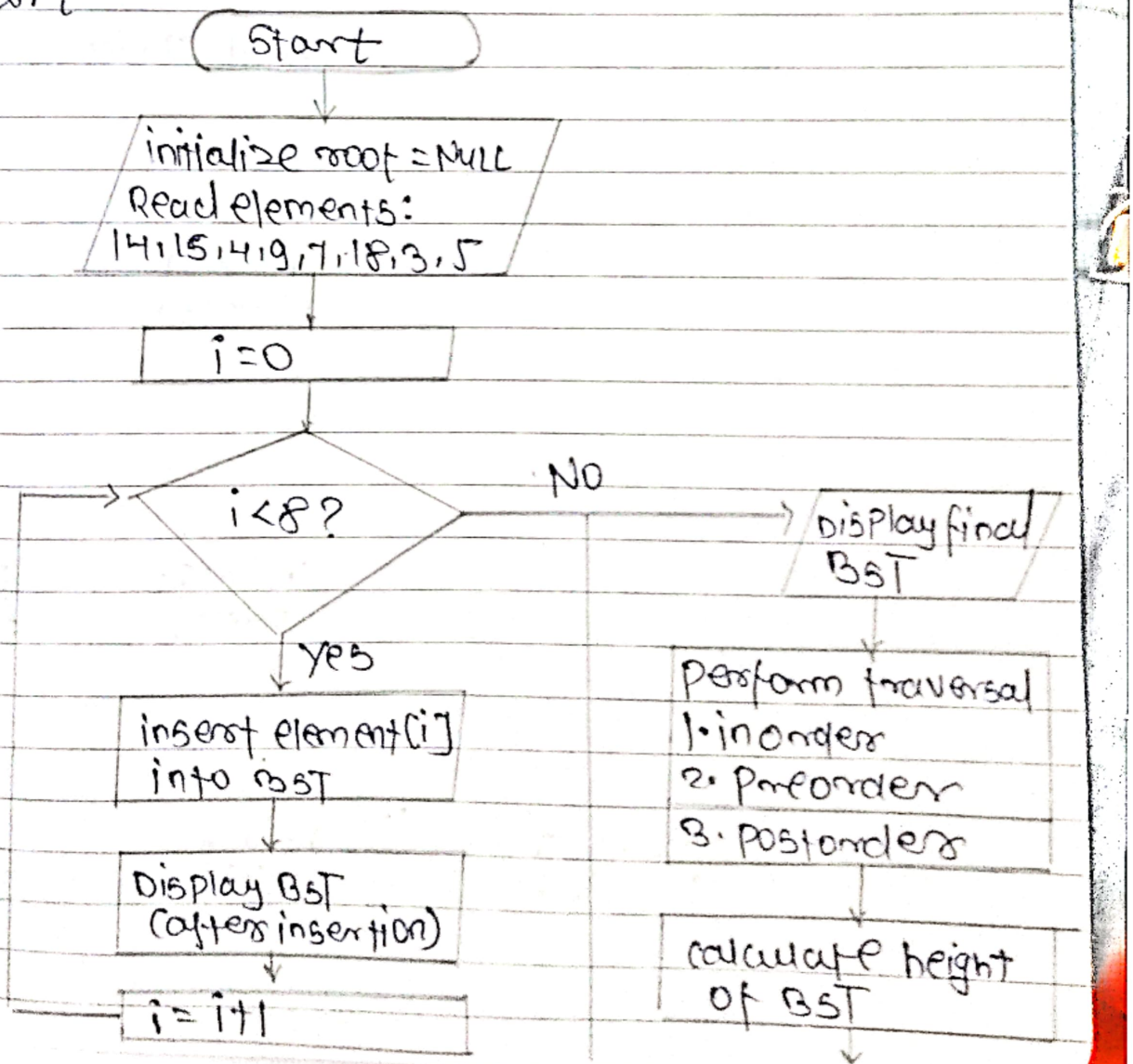
Step 6: Calculate the height of the final BST by finding the longest path from the root to a leaf node.
Step 7: count the leaf nodes, which are nodes having no left or right child.
Step 8: Ask the user to enter a value to search in the BST
- if the value exist display element found
- otherwise display element not found

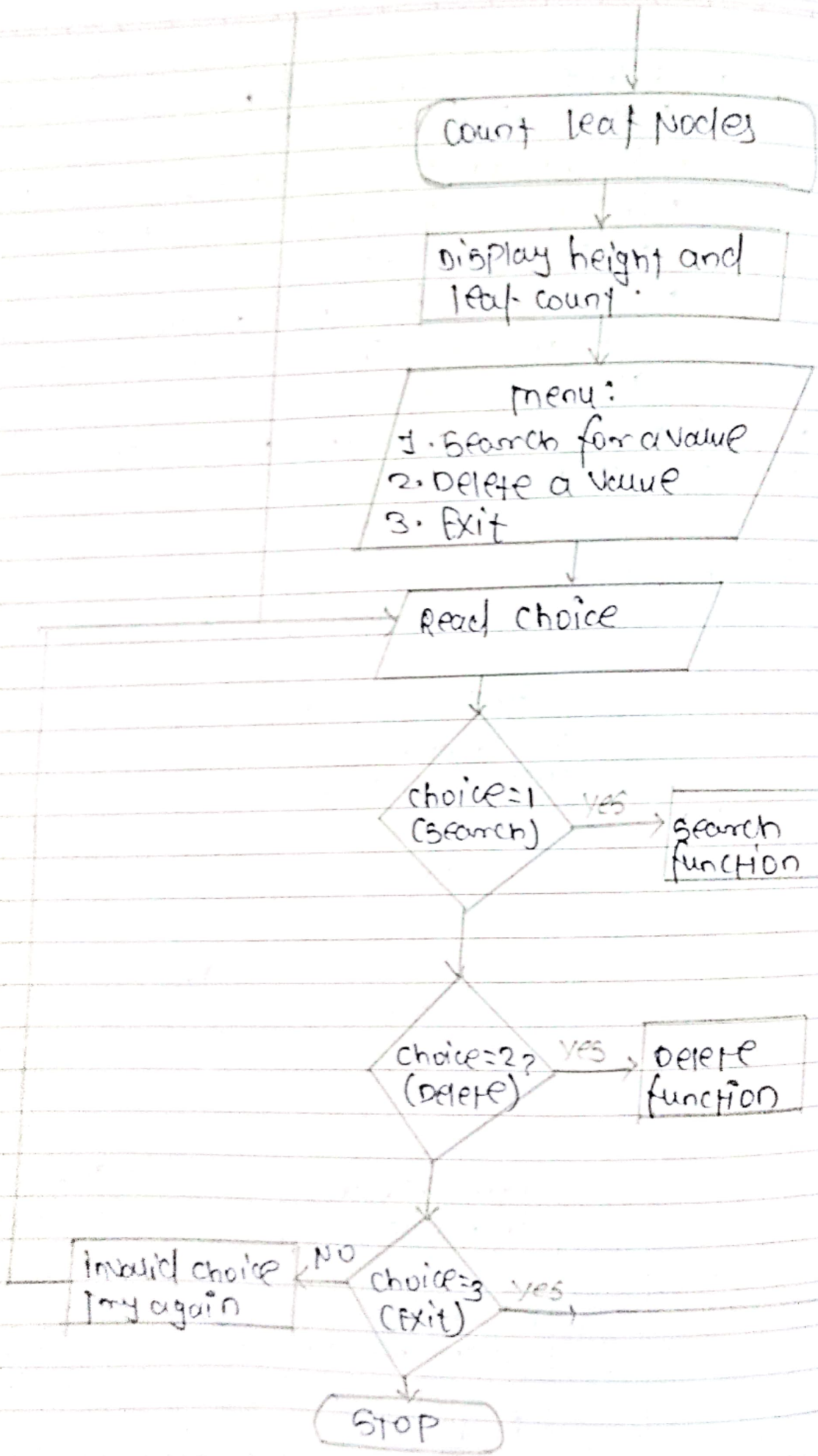
Step 9: Ask the user to enter a value to delete from the BST

Step 10: Delete the selected node according to the following cases:
- if the node is a leaf, remove it directly
- if the node has one child, replace it with its child.
- if the node has two children. replace it with its inorder successor and delete the successor

Step 11: Display the updated BST after deletion
Step 12: Stop.

flowchart




count leaf Nodes
display height and leaf count.
menu:
1. Search for a value
2. Delete a value
3. Exit
Read choice
choice=1 (Search)
yes
search function
choice=2? (Delete)
yes
delete function
choice=3 (Exit)
NO
Invalid choice Try again
yes
STOP